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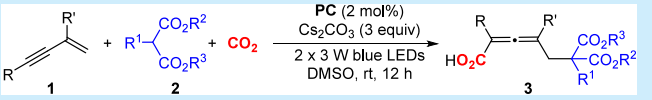
Visible-Light-Catalyzed Regioselective 1,4-Carbon/Carboxylation of 1,3-Enynes with Malonates and CO_2 toward 2,3-Allenoic Acids

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ABSTRACT: Difunctionalization of unsaturated carbon−carbon bonds with carbon dioxide (CO_2) provides expedient access to functionalized carboxylic acids. Compared to the well-developed carboxylation of alkenes or alkynes with CO_2, the catalytic carboxylation of 1,3-enynes has remained unknown. Herein, we report the first visible-light-promoted 1,4-carbon/carboxylation of 1,3-enynes with malonates and CO_2, leading to a series of valuable tetrasubstituted 2,3-allenoic acids with high efficiency. This new methodology features good functional group compatibility, ease of scale-up, and facile transformations. Mechanistic investigations indicate that successive single-electron transfer (SSET) reduction is involved to generate the key allenoic anion intermediate.



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Allenes are important structural motifs found in various natural products, bioactive pharmaceuticals, as well as functional molecules, such as lcariside, kumausellene, (+)-scorodonim, and (−)-panacene (Figure 1).[1] They can also serve as

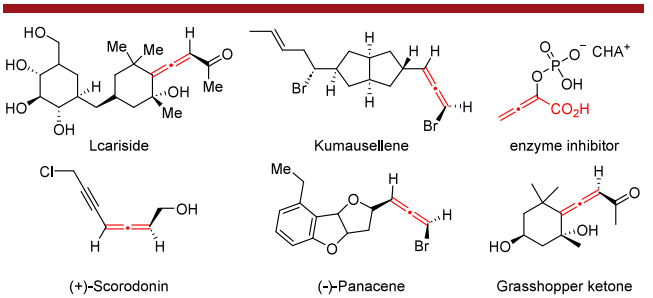
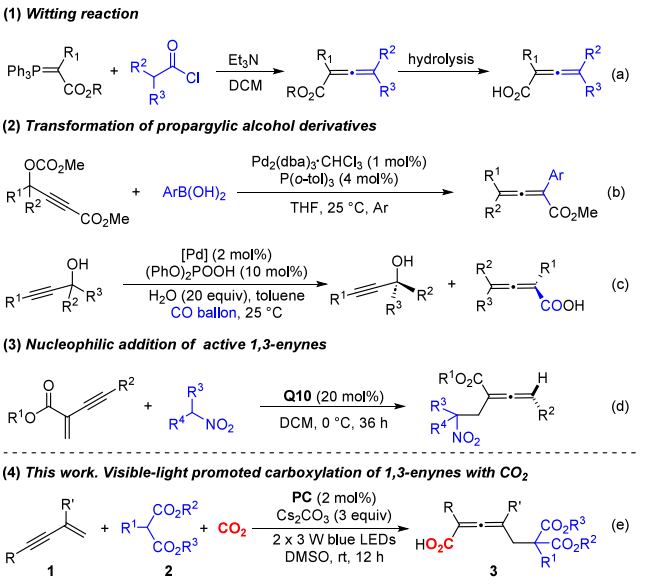


Figure 1. Representative examples of allenoic natural products and pharmaceuticals.

valuable intermediates and catalysts/ligands for a range of organic transformations.[2] Among them, 2,3-allenoic acids are of particular interest owing to their versatile reactivities,[3] such as the Morita−Baylis−Hillman reactions,[3d−f] rearrangements,[3g] cyclization reactions,[3h−j] and other transformations. Consequently, substantial efforts have been devoted to their synthesis.[4] Conventionally, allenoic acids could be synthesized via the Wittig reaction of ketenes with phosphorus ylides followed by hydrolysis, although this method suffered from a poor atom economy (Scheme 1a).[5] Recently, transition-metal-catalyzed functionalization of propargylic alcohols and their derivatives emerged as a practical method for the synthesis of 2,3-allenoates.[6] For example, Ma and co-workers pioneered an elegant palladium-catalyzed coupling reaction of 3-alkoxycarbonyl propargylic carbonates and boronic acids toward diverse tetrasubstituted 2,3-allenoates (Scheme 1b).[7] Furthermore, transition-metal-catalyzed carbonylation of propargylic alcohol

Scheme 1. Typical Routes for the Synthesis of 2,3-Allenoates



derivatives, using CO as the carbonyl source, provides an alternative efficient route to 2,3-allenoic acids (Scheme 1c).[8] Alternatively, organocatalytic nucleophilic addition of various nucleophiles to electron-deficient 1,3-conjugated enynes can

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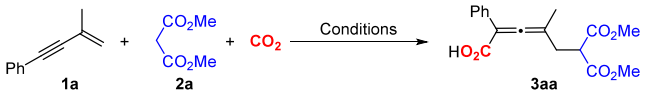
also furnish 2,3-allenoic acids with a good atom economy (Scheme 1d).[9] Despite these great achievements, the development of additional robust approaches for convenient access to 2,3-allenoic acids from readily available materials remains in high demand.

Carbon dioxide (CO_2) has emerged as an environmentally benign and renewable C1 feedstock in the synthesis of various value-added fine chemicals, owing to its natural abundance, cost-effectiveness, and low toxicity.[10] Among the diverse CO_2 transformations, the carboxylation reactions stand as one of the most convenient routes toward carboxylic acids.[11] In this field, carboxylation of unsaturated hydrocarbons with CO_2 is most attractive, as it avoids the use of halides or moisture-sensitive organometallic reagents.[12] For example, Martin and co-workers,[13] Yu and co-workers,[14] Wu and co-workers,[15] Xi and co-workers,[16] and we[17] have developed a series of practical visible-light-driven carboxylation reactions of alkenes with CO_2, enabling the synthesis of diverse functionalized and structurally tunable aliphatic carboxylic acids through judicious selection of radical precursors. Furthermore, transition-metal-catalyzed or base-promoted difunctionalization of alkynes with CO_2 also represents an effective route to furnish diverse acrylic acids.[18] For example, Wu and co-workers reported a novel route for the synthesis of lactones or acrylic acids by iridium/cobalt co-catalyzed hydro- or carbocarboxylation of alkynes with CO_2.[19] Additionally, Hong and Zhan independently reported the visible-light-promoted α-aminoalkyl- or phosphono-carboxylation of allenes with CO_2, leading to a series of β-functionalized allylic carboxylic acids with good efficiency.[20] However, to the best of our knowledge, the carboxylation of 1,3-enynes with CO_2 remains unreported to date. In view of the high value of allenoic acids and in continuation of our interest in the CO_2 chemistry,[21] we envisioned the photocatalytic carboxylation of 1,3-enynes with malonates and CO_2, leading to allenoic acids in one step (Scheme 1d).

Initially, (3-methylbut-3-en-1-yn-1-yl)benzene **1a** and dimethyl malonate **2a** were employed as the model substrates to test our hypothesis. When the reaction of **1a** with **2a** and CO_2 in DMSO was conducted under irradiation of 2 × 3 W blue LEDs with **PC1** as the catalyst and Cs_2CO_3 as the base. Delightedly, desired product **3aa** could be isolated in 33% yield (Table 1, entry 1). Encouraged by this preliminary result, we screened other parameters. First, some common photocatalysts were tested, and the results indicated that **PC4** gave the best result, affording **3aa** in 80% yield (entries 1−4). Additionally, the solvent also showed an obvious effect. In DMF, **3aa** was obtained in a 67% yield (entry 5). However, the reaction in NMP and MeCN afforded only **3aa** in 45 and 50% yields, respectively (entries 6 and 7). Common bases were also screened. Na_2CO_3, ^{t}BuOK, K_2CO_3, and K_3PO_4 all resulted in inferior results than Cs_2CO_3 (entries 8−11 vs 4). Furthermore, variation of the ratio of **1a**/**2a** to 1:2 or 2:1 showed no improvement (entries 12 and 13 vs 4). Lowering the catalyst loading to 1 mol % resulted in a diminished yield (59%, entry 14). Finally, control experiments demonstrated that the catalyst, light, base, and CO_2 were all essential to the success (entry 15).

Next, we investigated the reaction scope. First, a series of 1,3-enynes smoothly reacted with dimethyl malonate **2a** and CO_2 under the optimized conditions, affording the desired products with good efficiency (Scheme 2). It was found that the electronic nature of the substituents at the *para* position of the phenyl ring had no obvious effect. Substrates bearing

Table 1. Evaluation of Conditions[a]



entry	catalyst	base	solvent	**3aa** yield (%)
1	**PC1**	Cs_2CO_3	DMSO	33
2	**PC2**	Cs_2CO_3	DMSO	trace
3	**PC3**	Cs_2CO_3	DMSO	24
4	**PC4**	Cs_2CO_3	DMSO	80
5	**PC4**	Cs_2CO_3	DMF	67
6	**PC4**	Cs_2CO_3	NMP	45
7	**PC4**	Cs_2CO_3	MeCN	50
8	**PC4**	Na_2CO_3	DMSO	48
9	**PC4**	^{t}BuOK	DMSO	61
10	**PC4**	K_2CO_3	DMSO	66
11	**PC4**	K_3PO_4	DMSO	46
12[b]	**PC4**	Cs_2CO_3	DMSO	71
13[c]	**PC4**	Cs_2CO_3	DMSO	68
14[d]	**PC4**	Cs_2CO_3	DMSO	59
15[e]	**PC5**	Cs_2CO_3	DMSO	0

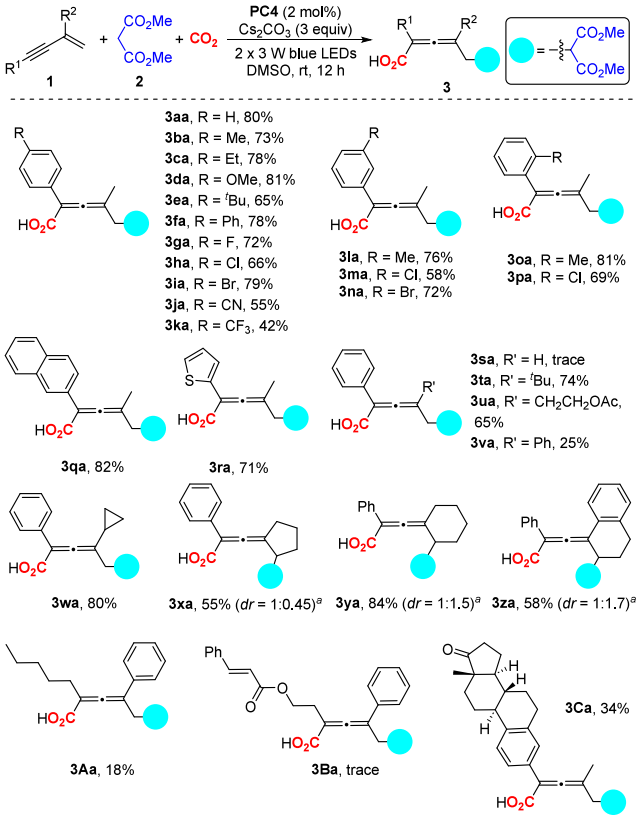
[a]Reaction conditions: **1a** (0.1 mmol), **2a** (0.11 mmol), base (0.3 mmol), CO_2 (1 atm), and solvent (1 mL), for 12 h, in a sealed Schlenk tube, irradiated by 2 × 3 W blue LEDs, unless otherwise noted. [b]**1a** (0.1 mmol) and **2a** (0.2 mmol). [c]**1a** (0.2 mmol) and **2a** (0.1 mmol). [d]Photocatalyst (1 mol %). [e]Without a catalyst, light, base, or CO_2; **PC1**, *fac*-Ir(ppy)$_3$; **PC2**, 4CzIPN; **PC3**, Ir[dF(CF$_3$)ppy]$_2$(dtbbpy)·PF$_6$; and **PC4**, Ir(ppy)$_2$(dtbbpy)·PF$_6$.

electron-donating groups, such as Me (**1b**), Et (**1c**), OMe (**1e**), ^{t}Bu (**1e**), and Ph (**1f**), and the one bearing electron-withdrawing groups, such as fluoro (**1g**), chloro (**1h**), bromo (**1i**), cyano (**1j**), and trifluoromethyl (**1k**), afforded **3ba**−**3ka** in 73−80% yields. Additionally, those with *m*-methyl (**1l**), *m*-chloro (**1m**), *m*-bromo (**1n**), *o*-methyl (**1o**), and *o*-chloro (**1p**) all worked well, leading to the desired products **3la**−**3pa** in 58−81% yields, indicating that steric hindrance had no obvious effect. Furthermore, 2-naphthyl- (**1q**) and 2-thienyl- (**1r**) substituted 1,3-enynes also afforded the desired products **3qa** and **3ra** in 82 and 71% yields, respectively. It is noteworthy that substitutions at the 3 position had a profound influence, as no desired product **3sa** could be detected when **1s** was employed. Delightedly, 1,3-enynes substituted with 3-^{t}Bu (**1t**), 3-CH_2CH_2OAc (**1u**), and cyclopropyl (**1w**) were employed, and the desired **3ta**, **3ua**, and **3wa** could be isolated in 74, 65, and 80% yields, respectively. However, 3-phenyl analogous **1v** could furnish only **3va** in 25% yield. Particularly, cyclic 1,3-enynes **1x**−**1z** were also good reaction candidates to deliver the desired products **3xa**−**3za**. Additionally, when non-1-en-3-yn-2-ylbenzene **1A** was employed, **3Aa** could be obtained in 18% yield. Unfortunately, cinnamoyl-containing substrate **1B** only resulted in a messy mixture with trace product detected. However, the one bearing an estrone motif **1C** was amenable to the reaction conditions, furnishing **3Ca** in 34% yield.

The scope of malonates was also investigated. A series of malonates bearing different substitutions reacted smoothly with **1a** and CO_2 (Scheme 3). For example, a range of malonate derivatives with different alkyl groups, including ethyl (**1b**), ^{n}Pr (**1c**), ^{i}Pr (**1d**), ^{n}Bu (**1e**), ^{t}Bu (**1f**), and benzyl (**1g**), were all compatible, affording the desired products **3ab**−**3ag** in moderate to good yields (58−76%). Furthermore, unsymmetrical malonates **2h** and **2i** also worked well to furnish the desired products **3ah** and **3ai** in 58 and 57% yields with a

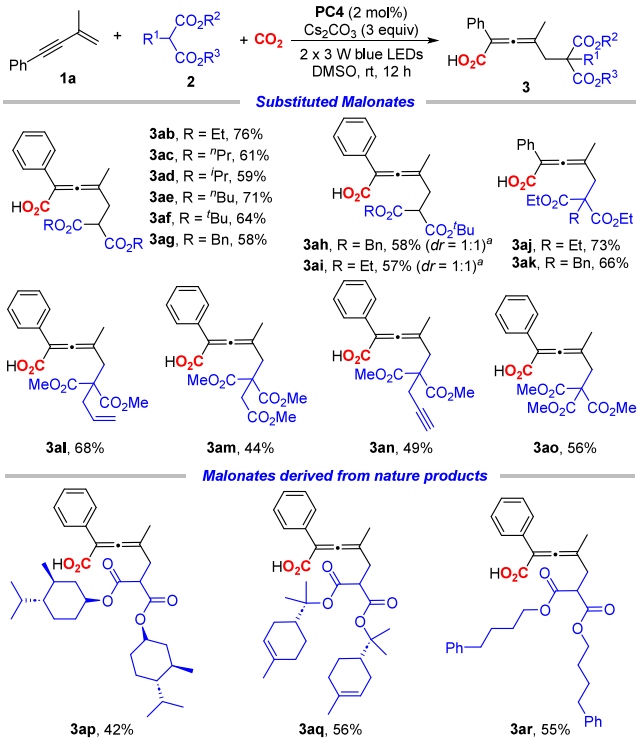
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Scheme 2. Scope of 1,3- Enynes



Reaction Conditions: **1** (0.2 mmol), **2a** (0.22 mmol), Ir(ppy)$_2$(dtbbpy)·PF$_6$ (0.004 mmol), Cs_2CO_3 (0.6 mmol, 3 equiv), DMSO (2 mL), and CO_2 (1 atm), for 12 h, in a sealed Schlenk tube, irradiated by 2 × 3 W blue LEDs, unless otherwise noted. [a]Determined by ^{1}H NMR.

Scheme 3. Scope of Malonates



Reaction Conditions: **1a** (0.2 mmol), **2** (0.22 mmol), Ir(ppy)$_2$(dtbbpy)·PF$_6$ (0.004 mmol), Cs_2CO_3 (0.6 mmol, 3 equiv), DMSO (2 mL), and CO_2 (1 atm), for 12 h, in a sealed Schlenk tube, irradiated by 2 × 3 W blue LEDs, unless otherwise noted. [a]Determined by ^{1}H NMR.

diastereomeric ratio (dr) of 1:1. Particularly, the substrate scope of malonates could be further expanded to substituted dimethyl ones. For example, those bearing ethyl (**2j**), benzyl (**2k**), allylic (**2l**), methyl ester (**2m**), propargyl (**2n**), and ester (**2o**) were all good reaction candidates to afford the desired products **3ja**−**3oa**. Notably, the unsaturated bonds of the dimethyl malonates (**2l** and **2n**) remained intact, providing opportunities for further derivatization. Additionally, malonates derived from natural alcohols were also compatible, yielding the desired 2,3-allenoic acids (**3ap**−**3ar**).

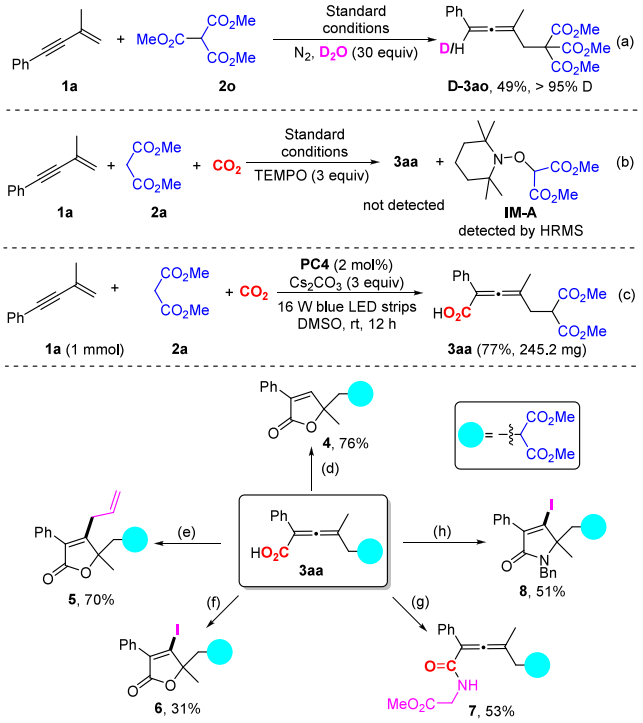
Then, some essential control experiments were conducted to investigate the possible reaction mechanism. As shown in Scheme 4, the reaction of **1a** and **2o** was conducted with the addition of an excess of D_2O (30 equiv). As excepted, the deuterated product **D-3ao** was isolated in 49% yield with more than 95% D at the allenyl position, indicating the generation of the allenyl carbanion (Scheme 4a). Upon addition of 3 equiv of TEMPO, the reaction was almost inhibited, with the detection of the radical adduct **IM-A** by HRMS (see Figure S4 of the Supporting Information for details), thus consistent with a radical pathway (Scheme 4b). Particularly, the luminescence quenching experiments indicated that this reaction was initiated by reductive quenching of photoexited Ir(ppy)$_2$(dtbbpy)·PF$_6$ by the cesium salt of malonates (see Figure S5 of the Supporting Information for details). Finally, a light on/off experiment indicated that continuous light irradiation was essential to the success (see Figure S6 of the Supporting Information for details).

To demonstrate the utility of this method, a 1 mmol reaction of **1a** with **2a** and CO_2 was conducted under the slightly modified conditions, leading to the desired product **3aa** in 77% yield (Scheme 4c). Then, further transformations of **3aa** were also conducted. First, a CuCl-catalyzed annulation of **3aa** proceeded smoothly, delivering the cycloaddition product **4** in 76% yield (Scheme 4d). Furthermore, a cyclization reaction of **3aa** catalyzed by $PdCl_2$ in the presence of allyl bromide afforded allyl furanone **5** in 70% yield (Scheme 4e). Treatment of **3aa** with *N*-iodosuccinimide (NIS) furnished iodo-lactone **6** in 31% yield (Scheme 4f). Additionally, the amination of **3aa** with methyl glycinate hydrochloride led to amide **7** in 53% yield (Scheme 4g). Alternatively, amination of **3aa** with benzylamine, followed by the treatment of the obtained amide with NIS, afforded iodo-lactam **8** in 51% yield (Scheme 4h).

Based on these results and the literature,[22] a mechanism is proposed in Scheme 5. Photocatalyst Ir(ppy)$_2$(dtbbpy)·PF$_6$ is excited by the irradiation of blue LEDs. Meanwhile, malonate **2** is deprotonated by Cs_2CO_3 to give its cesium salt **A**. Then, reductive quenching of photoexcited Ir(ppy)$_2$(dtbbpy)·PF$_6$ by intermediate **A** forms radical species **B** and reduced photocatalyst PC$^{\bullet -}$. Then, radical addition of **B** to 1,3-enyne **1** affords radical **C**. SET reduction of **C** by PC$^{\bullet -}$ forms its anion counterpart **D** and regenerates Ir(ppy)$_2$(dtbbpy)·PF$_6$. Intermediate **C** undergoes tautomerization to form allenoic carbanion **D**, which then undergoes nucleophilic addition to

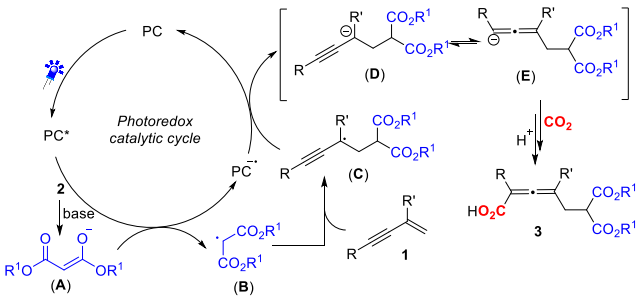
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Scheme 4. Mechanism Studies and Derivatizations



Reaction Conditions: (d) CuCl (4 mol %), MeOH, 60 °C (Oil Bath), 1 h; (e) $PdCl_2$ (2 mol %), Allyl Bromide (6 equiv), 50 °C (Oil Bath), 24 h; (f) NIS (1.2 equiv), MeCN; (g) EDCI·HCl (1.3 equiv), Et_3N (1.3 equiv), DMAP (10 mol %), Glycine Methyl Ester Hydrochloride (1.5 equiv); and (h) Amidation with $BnNH_2$ as Described in Reaction g, Followed by Reaction with NIS.

Scheme 5. Proposed Mechanism



CO_2. Subsequent acidification with aqueous HCl yields carboxylic acid **3**.

In conclusion, we have developed the first visible-light-promoted 1,4-carbon/carboxylation of 1,3-enynes with malonates and CO_2, leading to a series of valuable tetrasubstituted 2,3-allenoic acids in good yields. This new methodology features good functional group tolerance, ease of scale-up, and facile transformations. Mechanistic investigations indicate that successive single-electron transfer (SSET) reduction is likely involved, with the allenoic anion serving as a key intermediate.

■ ASSOCIATED CONTENT

Data Availability Statement

The data underlying this study are available in the published article and its Supporting Information.

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.orglett.5c02893.

Experimental procedures, product characterization data, and spectra (PDF)

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Author Contributions

This work was conducted by Aolong Zhang, written by Song Sun, and polished by Jianwei Sun. Han Yang and Yang Yao helped with the sample testing. All authors have given approval to the final version of the manuscript.

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Notes

The authors declare no competing financial interest.

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